

These notebooks are directionally strong, but not validation-ready without additional evidence, documentation, and independent testing.

Overall Validator Opinion

The notebook pipeline is logically structured:

Policy processing → Borrower processing → Merge/target creation → Data validation → Model development → Metric thresholds

That is a good structure. However, from a validation perspective, I would not approve yet until the following items are tested and documented.

Key Strengths

1. Clear pipeline separation

- Policy data, borrower data, merge, validation, model development, and thresholds are separated.

2. Version history exists

- Several notebooks include version notes, which is good for governance.

3. Data validation notebook is useful

- It includes profiling, portfolio summary, feature profiles, drift/stability analysis, and target relationship review.

4. Model development includes validation outputs

- Cross-validation, AUC/log loss/Brier score, segment analysis, SHAP, risk ranking, and decile analysis are present.

5. Threshold notebook compares SMI2 vs SMI3

- This is useful challenger/benchmark evidence.

Main Validation Concerns

1. Row-count reconciliation needs to be formalized

I see row counts printed in notebooks, but as a validator I would require a formal table:

Step	Input Rows	Output Rows	Difference	Reason
Raw policy load	X	X	0	Source load
Drop missing application number	X	Y	Z	Invalid ID
Drop duplicates	Y	Z	N	Keep last
Merge borrower	Z	A	B	Left join

Finding risk: Medium

Why: Without this, it is hard to prove no unintended data loss or duplication.

2. Duplicate treatment needs stronger justification

In the policy notebook, duplicates are dropped using APPLICATION_NUMBER, keeping the last record.

As validator, I would ask:

- Why keep the last record?
- Is the notebook sorted before dropping?
- Does “last” mean latest effective date, latest update timestamp, or just file order?
- Are duplicate cases reviewed?

Potential finding: Medium to High if not justified.

3. Target definition is critical and needs deep validation

The merge notebook creates target logic based on:

- Declined status
- Claim indicator
- 90-day delinquency indicator

As validator, this is the most important area.

I would validate:

- Are declined loans truly treated as bad/event?

- Are claim and delinquency flags mutually exclusive?
- Are “good declines” filtered correctly?
- Is there any future information leakage?
- Is the target aligned with intended model use?

Potential finding: High if target logic is not fully justified.

4. Missing-value treatment needs business rationale

Some features are filled with zero.

That may be valid, but it must be proven.

Ask:

- Does missing mean truly zero?
- Or does missing mean unknown/not reported?
- Should missingness be a separate indicator?
- Are default rates different for missing vs non-missing?

Potential finding: Medium.

5. Credit score processing needs careful validation

The borrower notebook does credit score cleaning, clipping, binning, and aggregation.

I would test:

- Score range validity
- Missing score treatment
- Use of zero as valid score
- Bureau hierarchy logic
- Average/max score aggregation
- Binning reasonableness

Key concern: Credit score is likely a top predictor, so any processing error can materially affect the model.

6. Merge logic needs independent testing

The merge notebook uses left join from policy to borrower.

I would validate:

- Is APPLICATION_NUMBER unique before merge?
- Does the merge create duplicate rows?
- What percentage of policies fail to match borrower data?
- Are unmatched records materially different?
- Are target/event rates different for matched vs unmatched?

Potential finding: High if merge creates duplication or selection bias.

7. Feature clipping needs justification

The notebooks clip certain features.

As validator, I would ask:

- What are the clipping thresholds?
- Are they based on business rules, percentiles, or modeler judgment?
- How many records are affected?
- Does clipping change rank ordering materially?

Potential finding: Medium.

8. Model methodology appears reasonable, but needs more support

The development notebook appears to use XGBoost/model trainer logic, cross-validation, SHAP, risk ranking, and segment analysis.

That is acceptable, but I would require:

- Why this algorithm was selected

- Challenger model comparison
- Hyperparameter tuning rationale
- Overfitting assessment
- Out-of-time validation
- Calibration analysis

Potential finding: Medium.

9. Performance looks strong, but recall may be weak

One output shows approximately:

- AUC around 0.905
- Accuracy around 0.954
- Recall around 0.281
- Precision around 0.675

That means the model ranks risk well, but may miss many actual events depending on threshold.

As validator, I would not rely on accuracy because the event rate is imbalanced.

I would focus on:

- AUC/Gini
- KS
- Recall at business threshold
- Lift by decile
- Calibration
- Loss ratio ranking

10. Threshold notebook needs business approval evidence

The threshold notebook compares SMI2, SMI3 funded, and SMI3 all-data outputs.

As validator, I would ask:

- Why were these thresholds selected?
- Are thresholds stable by region/book year?
- What is the business impact of threshold changes?
- Are there override rules?
- Is there monitoring if score distribution shifts?

Potential finding: Medium.